SQL-With-Baraa-Section-29-Indexes-UNIQUE-vs-FILTERED

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Unique Index

[Ensures no duplicate values exist in specific column.

Bnfits

1. Improve data integrity
2. Improve query performance.

Performance

Writing → Slower
Reading → Faster
> Than non-unique case.

Index

```
CREATE [UNIQUE] [CLUSTERED/NONCLUSTERED] [COLUMNSTORE]

INDEX idx_<table_name>_<col_name>
ON    <table_name> (col,...)
```

If unique is specified, duplicates are not allowed.

otherwise, columns with duplicates are allowed.

If a column is used in unique clustered or Nonclustered indexing, Duplicate values cannot be inserted into the column

Filtered Index

An index → That includes rows only when

↓

Meeting specific conditions

Create b-trees using rows that meet the condition [Probably only for NONCLUSTERED CASE?]

Benefit:

1. Targeted optimization.

optimize rows that are heavily used.

2. Storage reduction. Duh.

Syntax

```
CREATE [UNIQUE] [NONCLUSTERED]
INDEX  idx_table_column
ON     table (column)
WHERE  [condition]
```

SQL Server

- ✗ CLUSTERED
- ✗ COLUMNSTORE

→ ✗ CLUSTERED COLUMNSTORE

When to use which indexing

HEAP

Fast writes

Fast writes

Clustered Index

For Primary keys

OLTP systems

Columnstore Index

OLAP Systems.

Non-clustered Index

For not primary keys

(Foreign keys, Joins, Etc)

Filtered Index

Target subset of data

Optimize size and performance.

Unique

Enforce uniqueness.

Improve query speed.

When To Use

HEAP

Fast **Inserts**
(For Staging Tables)

Clustered Index

OLTP

For **Primary keys**
If not, then for date columns

Columnstore Index

OLAP

For **Analytical** Queries
Reduce Size of Large Table

Non-Clustered Index

For **non-PK** columns
(Foreign keys, Joins, and Filters)

Filetered Index

Target **Subset** of Data
Reduce Size of Index

Unique Index

Enforce **Uniqueness**
Improve Query Speed